

UGED1111 Logic

Basic Concepts II

Evaluate an argument

Standard

- ❑ Whether the premise(s) support the conclusion
- ❑ Whether the premise(s) are true

- Different standard of Evaluation for Deductive Argument and Inductive argument
 - Validity and Soundness
 - Strength and Cogency

Types of Argument

Deductive argument (演繹論証)

1. 所有中大學生都長得好看。
 2. 小明是中大學生。
-
3. 小明長得好看。

Inductive argument (歸納論証)

1. 96 % 的中大學生都長得好看。
 2. 小明是中大學生。
-
3. 小明長得好看。

Types of Argument

◆ A **deductive argument** (演繹論証)

incorporates the claim that it is **impossible** for the *conclusion to be false given that the premises are true*.

◆ An **inductive argument** (歸納論証)

incorporates the claim that it is **improbable** that the conclusion be false given that the premises are true.

Deductive Argument

Example 1:

1. 中年的定義是人均壽命的一半。
 2. 人均壽命百二歲。
-
3. 六十歲是中年。

Example 2:

1. 所有貓都有嘴。
 2. Hello Kitty 是貓。
-
3. Hello Kitty 有嘴。



Inductive Argument

Example 1.

1. 十個光頭九個富。
 2. 劉小煒是光頭的。
-

3. 劉小煒是富翁。

Example 2

1. 今天亞視存在。
2. 昨天亞視存在。

...

3. 亞視會永遠都存在。



Concepts of Possibility



What does it mean when we say something is “possible” or “impossible”?

“It is impossible for us to travel to Jupiter”

當男神/女神跟你說「我們不可能」

Concepts of Possibility



- **Technological Possibility**

- ❑ **Not exceed the limitation of current technology**

- **Empirical Possibility**

- ❑ **Not against any laws of nature**

- **Logical Possibility**

- ❑ **Not against fundamental laws of logic**

Concepts of Possibility

Technological Possibility

-Not exceed the limitation of
current technology

Example:

It is impossible for us to
travel to Jupiter.



Concepts of Possibility

Empirical Possibility

--Not against any laws of nature

e.g. 永動機



Concepts of Possibility

Logical Possibility

- Not against fundamental laws of logic
- Not involve any contradictions

Example

- Rounded-square

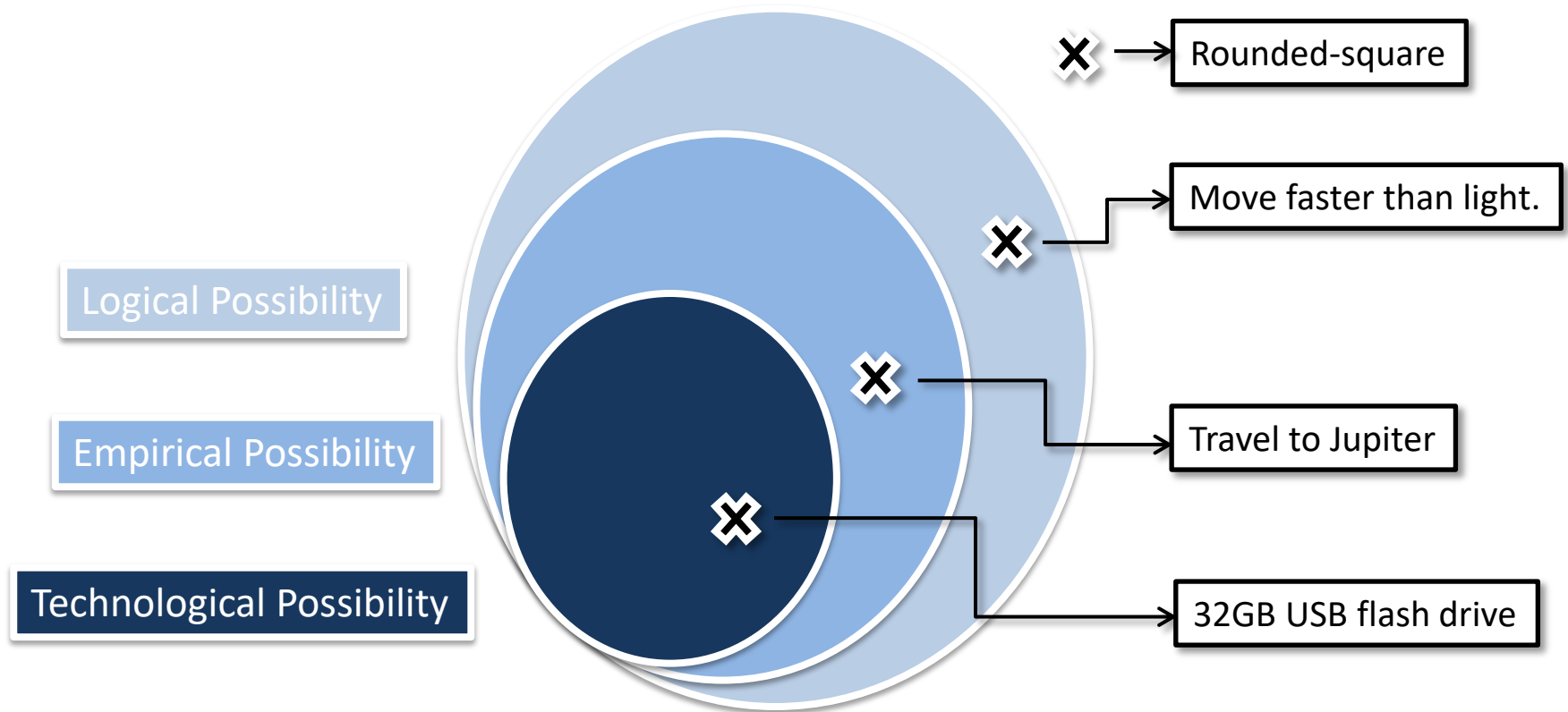


Concepts of Possibility

1. 如果上帝不可能造出自己舉不起來的石頭，上帝不是全能的。
2. 如果上帝可能造出自己舉不起來的石頭，上帝不是全能的。
3. 上帝可能造出自己舉不起來的石頭或上帝不可能造出自己舉不起來的石頭。

上帝不是全能的。

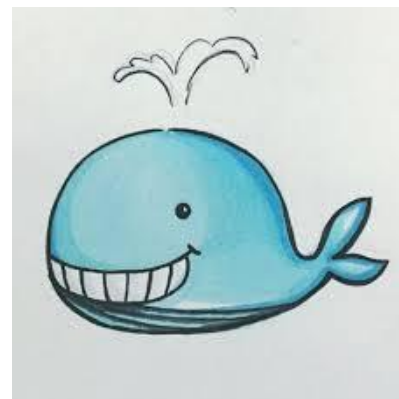
Concepts of Possibility



Question

1. 所有的鯨魚都需要水(T)
2. 所有的魚都需要水。(T)

3. 鯨魚是魚。(F)



Good argument or bad argument ?

Validity

Valid argument(對確論證):

◆ It is impossible that all its premises are true and the conclusion is false.

Conversely,

Invalid argument:

◆ It is possible that all the premises are true and the conclusion is false.

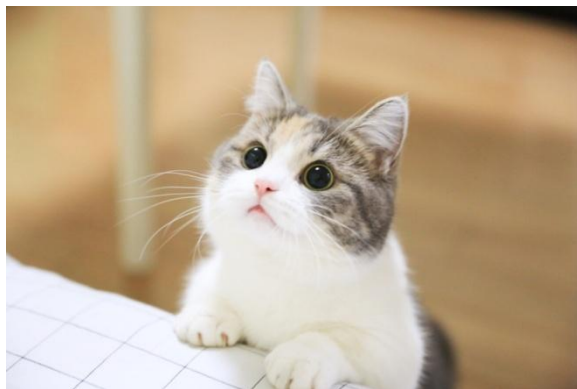
Case 1a

1. 如果你是中大學生，則你有中大學生証。(T)
 2. 你是中大學生。(T)
 3. 你有中大學生証。(T)
-

Valid

Case 1b

1. 所有的哺乳類都是需要空氣生存的生物。(T)
 2. 貓是需要空氣生存生物。(T)
 3. 貓是哺乳類。(T)
-



Invalid

Case 2

1. 所有的貓都是哺乳類(T)
2. 所有的兔都是哺乳類。(T)

3. 所有貓是兔。(F)

Invalid



Case 3a

1. 所有長髮的都是可愛的。(F)
2. 所有可愛的都是生理男性。(F)
3. 圖中的人是長髮的。 (T)

4. 圖中的人是生理男性。(T)



Valid

Case 3b

1. 所有不達156 cm 高的人都是小孩。 (F)
 2. 夏綠蒂是小孩。 (T)
-
3. 夏綠蒂不達156cm 高。 (T)

Invalid



Case 4a

1. 所有靚仔都有老婆。 (F)

2. 八丙金有老婆。 (T)

3. 八丙金是靚仔。 (F)



Invalid

Case 4b

1. 所有不達156 cm 高的人都是小孩。(F)
 2. 陳希敏不達156 cm 高。(T)
-
3. 陳希敏是小孩。(F)

Valid

Remarks

*An argument is valid or not **does not depend** on the actual truth value of the **premises** and the **conclusion**.*

Premises	Conclusion	Validity
T	T	?
T	F	Invalid
F	T	?
F	F	?

Special Case 1

1. 陳小波是人。
 2. 陳小波不是人。
-
3. 鯨魚是魚。

Valid

Premise 1 and Premise 2 is inconsistent. They cannot be both true.

Therefore, it is logically impossible that the conclusion is false and all the premises are true

Special Case 2

1. The government knows the need of all citizens.
 2. The air pollution problem in Hong Kong is not serious.
-
3. Mimi has a tail or Mimi does not have a tail.

Valid

The conclusion is necessarily true.

Therefore, it is logically impossible that the conclusion false and all premises is true

Vacuous Validity

- Arguments with **inconsistent premises/ necessarily false premise(s)**
- Arguments with a **necessarily true conclusion**
- counter-intuitive?

Any better accounts ?

Def_{new}: For any valid argument, it is impossible that all the premises are true and the conclusion is false; and

- 1. it is not the case that (a) the premises are necessarily false or inconsistent , and**
- 2. it is not the case that (b) the conclusion is necessarily true.**

Problems of the suggested definition

- The concept of validity no longer apply on the mathematical proof / proof of theorem in Logic.
- Against the second additional constraint

Problems of the suggested definition

1. Hello Kitty is a cat with a month.
2. Hello Kitty does not have a month. (inconsistent Premises)
3. Hello Kitty is a cat.

1. 所有可愛的貓都是貓。
2. 不是貓的都不是可愛的貓。 (Necessarily True)

It will eliminate some cases which we consider as “good” inference.

Truth Preserving

We will keep the original definition

- It captures an important feature of good deductive argument – truth preserving
- When we say the argument is invalid, what we means is that it cannot guarantee the truth of the conclusion.

**Remarks: An argument is either valid or invalid.
We will not say an argument is N % valid.**

Discussion

1. 獲得我最喜愛男歌手的人都是唱歌好聽的男歌手。
2. 古天樂獲得我最喜愛男歌手。
3. 古天樂是唱歌好聽的男歌手。



Soundness

$$\text{Sound Argument} = \text{Valid Argument} + \text{All True Premises}$$

Sound argument:

- A deductive argument that is valid and has all true premises.

Examples

Example 1

1. 蘇格拉底是人類。 (T)
 2. 所有人類都會死亡。 (T)
-
3. 蘇格拉底會死亡。 (T)

Example 2

1. Putin is a bear. (F)
 2. All Bears are mortal. (T)
-
3. Putin is mortal. (T)

Reference

- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- Lau, J. (2011). *An Introduction to Critical Thinking and Creativity : Think More, Think Better*. Hoboken, N.J.: Wiley.